

Using Machine Learning to Predict Alcohol Consumption Intensity

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Introduction:

Our research question uses All of Us research data and machine learning to predict AUDIT-C scores for patients' alcohol consumption intensity, based on their self-reported survey answers. We will test a variety of models and determine the most significant factors in predicting AUDIT-C scores.

Excessive alcohol consumption is a common problem in America, with over 178,000 people a year dying from excessive drinking (Centers for Disease Control and Prevention [CDC], 2024). Excessive drinking is dangerous and has many long-term risk factors, including chronic disease, injury, and premature mortality. To help reduce the problem of drinking in America, it is important to be able to identify which patients are at high risk of developing a drinking disorder, so proper precautions, and in some cases, treatment can be administered to combat alcoholism.

Methods:

We defined our study cohort as patients aged 22-65 who had completed all three of the following alcohol consumption survey questions so that we could calculate their AUDIT-C scores: 1) How often did you have a drink containing alcohol in the past year? 2) On a typical day when you drink, how many drinks do you have? 3) How often did you have six or more drinks on one occasion in the past year? We chose 22 as the bottom of our age range, as patients were asked if they had consumed alcohol in the past year, and we wanted them to have been of legal drinking age in the United States in the previous year. We chose 65 as the top of the age range, as drinking habits change as people get older, and we wanted our models to generalize to younger patients. There were 262,087 patients in our initial cohort.

We calculated the AUDIT-C scores, on a scale of 0-12, based on the sum of their scored response to each question. 1) Never = 0, Monthly or less = 1, Two to four times a month = 2, Two to three times a week = 3, Four or more times a week = 4. 2) 1 or 2 = 0, 3 or 4 = 1, 5 or 6 = 2, 7 to 9 = 3, 10 or more = 4. 3) Never in the last year = 0, Less than monthly = 1, Monthly = 2, Weekly = 3, Daily or almost daily = 4.

We defined the index date as the latest date and time at which a patient completed one of the three alcohol consumption survey questions. We then add one day to each patient's index date to allow other survey questions completed by patients on the same day at a later time to be included in our analysis. Our observation window is two years before a patient's index date. All predictors for our patients must fall between their observation window and index date, to ensure that predictors are relevant to the AUDIT-C score when it was calculated and were not developed after the score was computed. We also wanted them to be timely, so that a patient's responses are likely similar on the day they filled out the alcohol survey questions and have not changed before the index date.

We included demographic, lab measurements, survey, conditions, and medication use data in our predictors. For demographic data, we create a race 2 variable, which had the value Hispanic if a person's Ethnicity was Hispanic, and recorded their race if a patient was not Hispanic. We only included non-Hispanic patients with the following races: "White", "African-American", and "Asian". This included the majority of our patients and made results more interpretable than including the other excluded races of "Another single population" and "None of these". We also included sex assigned at birth and age, calculated by the difference in time from a person's index date and their date of birth, to get their age when the AUDIT-C score was created. Sex assigned at birth was converted to a boolean with 1 being Male and 0 being Female.

We included three lab measurements as predictors: BMI, Aspartate aminotransferase (AST), and Alanine aminotransferase (ALT). Abnormal values in all three of these measurements have been correlated with alcoholism. BMI can be high due to an increase in weight from alcohol use, and high AST and ALT values are key indicators of alcohol-related liver damage. Once again, all of these measurements fell between the index date and observation window for each patient, and we used the most recent values for each patient, for each measurement in that window. We observed several unreasonable lab measurements that were likely errors. We removed these unreasonable measurements from our data. The boundaries we used for these unreasonable measurements were BMI less than 10 and 80 and above, as well as AST/ALT less than 5 and 1500 and above.

We included the following survey data as predictors: education level, employment status, household income, self-reported physical health, self-reported mental health, self-reported social health, marital status, recreational drug use, and current smoking status. These predictors were all generated from self-reported survey questions. All of these survey questions were completed between our index date and observation window for each patient. If a patient filled out the same survey multiple times in this window, the latest survey value was used. We also filtered out all survey responses that didn't give an interpretable answer, such as "Prefer Not to Answer" or "Skip". All of these predictors were categorical variables with 3-5 possible responses, except for household income, which had 10, and employment status, which had 2 categories and was recoded as a binary variable with 1 representing employment and 0 representing unemployment. All three of the categorical health score responses were converted to ordinal values using the following methodology: "Poor" = 0, "Fair" = 1, "Good" = 2, "Very Good" = 3, and "Excellent" = 4. Education level was converted to a boolean representing whether a person had any college education. The responses "college graduate or advanced degree" and "college one to three years" were given a value of 1, while the responses "twelve or GED" and "less than a high school degree or equivalent" were given a value of 0. Current smoker status was also a boolean with 1 representing a current smoker and 0 representing not a current smoker. It was calculated by patients who had smoked a traditional cigar, cigarillo, or filtered cigar, even one or two puffs, and now smoke a traditional cigar, cigarillo, or filtered cigar.

We used four different alcohol-related comorbidities as predictors: anxiety, depression, hypertension, and liver disease. People who have substance abuse problems frequently have underlying conditions such as anxiety and depression that lead them to drink. Patients who excessively drink often have high blood pressure and hypertension. Similarly, patients with liver

disease often have damaged livers due to excessive drinking. For each patient, we made sure that their condition start time was between the patient's index date and observation window, to ensure the disease was current. We also only included the most recent form of each disease in that window for that patient. For example, if a patient had both PTSD and general depression during the observation window, we only kept the most recent form of depression. We then recoded all of these conditions as binary variables, so if a person has some form of that comorbidity, we assign them a 1 in that column, and if not, we assign them a 0.

We included three different medications prescribed for alcohol use disorder in our predictors. They were the three main FDA-approved drugs to treat it: naltrexone, acamprosate, and disulfiram. We filtered for medication use that started between a patient's index date and observation window. Since there was a small number of patients taking each drug, and each drug serves similar purposes, to treat patients with alcoholism, we combined their use into one feature. We created a binary variable where if a patient is on at least one alcoholism medication during their observation window, we assign them a 1; if not, we assign them a 0.

Finally, we used an age and sex at birth interaction to see if alcohol consumption differs for each sex as a person ages.

For all categorical variables that were not converted to binary or ordinal variables, we used one-hot encoding. For each one-hot encoded feature, we dropped the first category of each categorical variable to prevent multicollinearity from disrupting the linear models.

We currently have 21 different predictors for AUDIT-C scores (not including extra columns generated from one-hot encoding), and only included patients in our cohort with valid values (non-missing) for all of these predictors. We ended up with 31,174 patients in our cohort fulfilling that requirement and having valid data for all predictors. This is more than enough data to proceed and create our machine learning models.

In Figure 1, you can see the number of patients remaining as we merge each table together to get from the initial cohort (262,087 patients) to the final cohort (31,174 patients), where all patients have values for all the covariates. Notably, the measurement data was the sparsest and the type of data we had the least of. Also, we had condition and medication data for all patients; if we didn't have records of a condition or medication, we assumed they didn't have that condition or weren't taking that medication.

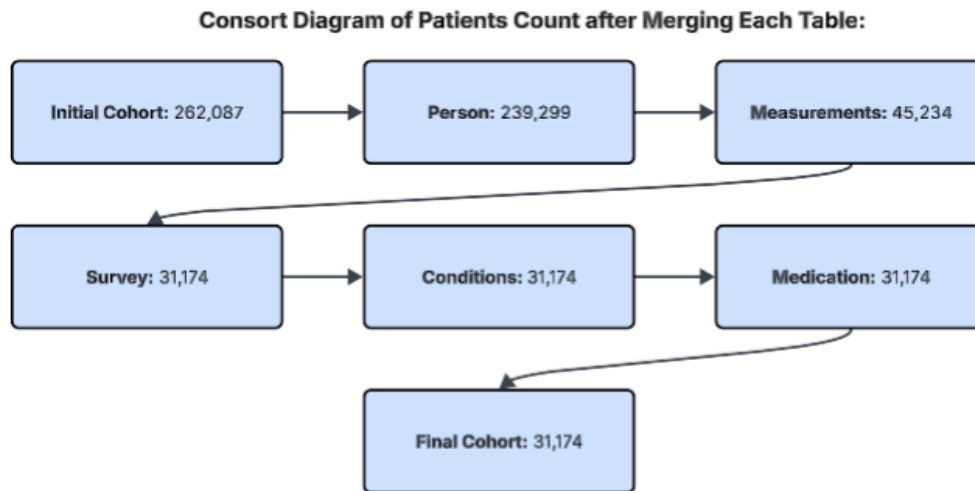


Figure 1: Consort Diagram of Patient Count after Merging Each Table

Results:

The final cohort size was 31,174 patients, and key demographics for the final cohort can be seen in Figures 2-4. Notably, the final cohort is majority female, with over 65.3% patients being female. This is likely due to females volunteering to be a part of the All of Us program more than males. Additionally, White people make up the majority of the race distribution at 60.3%, followed by Hispanics at 20.0%, Black people at 16.2%, and Asians, the biggest minority in the cohort, at 3.6%. The age distribution was pretty constant except for fewer participants on the younger and older ends. 83.3% of patients were in the middle ages of 30-59. The average and median ages were 42.9 and 44.0, respectively.

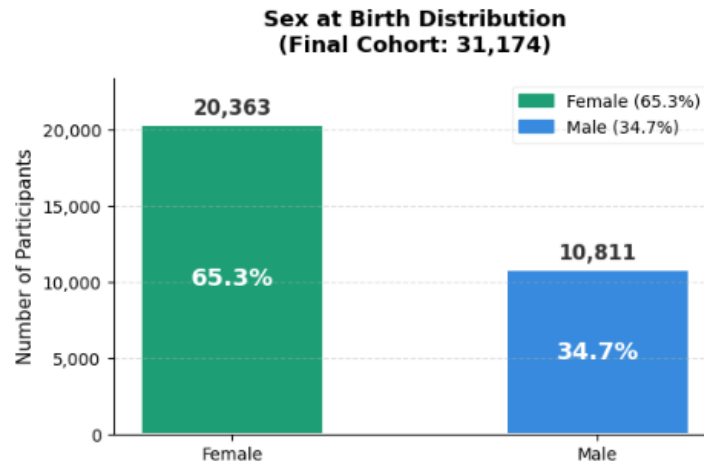


Figure 2: Sex at Birth Distribution for Final Cohort

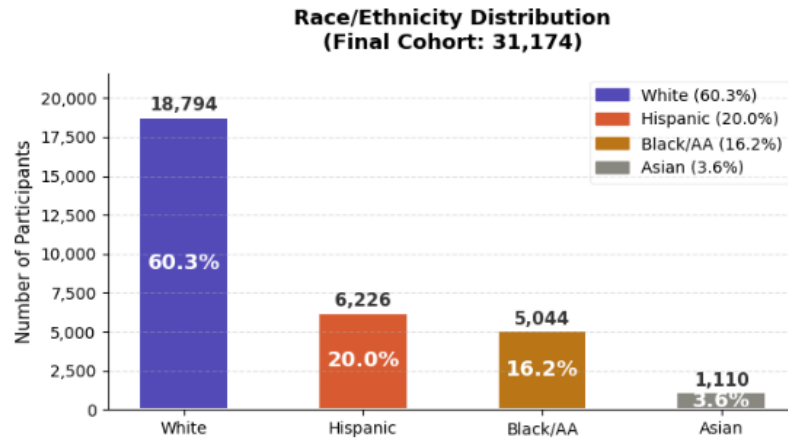


Figure 3: Race/Ethnicity Distribution for Final Cohort

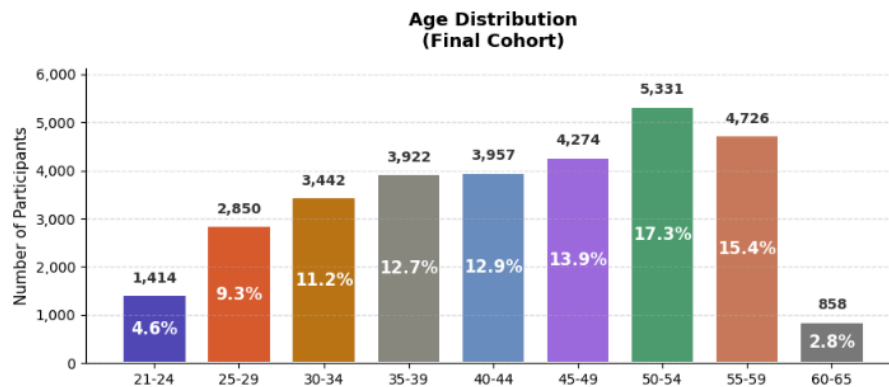


Figure 4: Age Distribution for Final Cohort

The distribution of AUDIT-C scores is right-skewed and can be seen in Figure 5. The mean was 3.08, the standard deviation was 2.21, and the median was 2. Most people had lower drinking scores and thus were at lower risk of being hazardous drinkers. Patients with high scores were quite rare. However, AUDIT-C scores are self-reported, so they may not be representative of patients' real drinking behaviors, as some patients may change their answers out of embarrassment or social pressure. No patient had a score of 0, which is otherwise a viable score, because to be in the cohort, they must answer all three alcohol consumption questions. Patients who answered never drinking in the past year for the first question were not asked the other two survey questions, making them ineligible to be in the cohort, resulting in no patients with a score of 0.

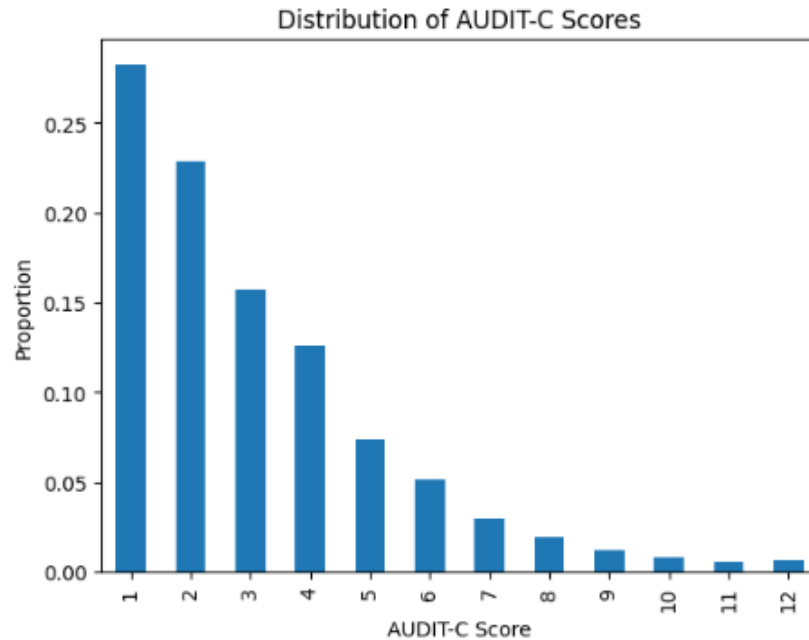


Figure 5: Distribution of AUDIT-C Scores in the Final Cohort

Figure 6 shows a heatmap of the correlation between all numeric features in the final cohort. The more red the square, the stronger the correlation. The correlation between features and AUDIT-C scores can be seen in the first column. Notably, none of the predictors have an incredibly strong correlation with AUDIT-C scores; the highest is the boolean for sex at birth, with male being a value of 1, which has a correlation of 0.22. The only other predictors with an r-squared around 0.1 are: the current smoker boolean at 0.17, alcoholism medication at 0.13, BMI at -0.11, AST at 0.11, and the boolean for some college education at 0.10. There doesn't appear to be a strong linear relationship between any of our numeric predictors and AUDIT-C scores. Additionally, some of our features appear to be moderately to heavily correlated with each other, reducing the importance of having both features in our model. These heavily correlated terms are: AST and ALT at 0.77, anxiety and depression at 0.48, and physical, mental, and social health at 0.49, 0.47, and 0.64.

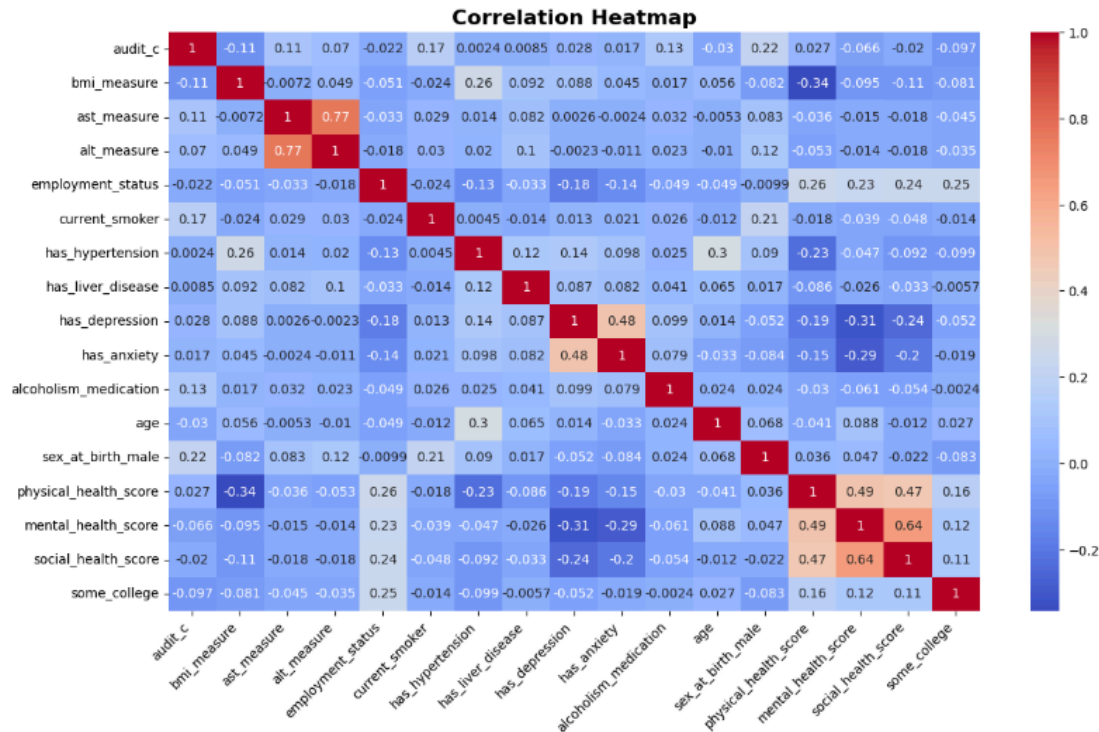


Figure 6: Correlation Heatmap for Numeric Predictors of AUDIT-C scores

Most prior research on predicting AUDIT-C scores uses classification methods to group patients from severe risk to low risk based on a range of scores. While we could have continued this trend and used classification models in our research, we aimed to explore a more novel approach and see how regression models performed. We treated the AUDIT-C score as a continuous variable and tested the following regression models: Linear Regression, Linear Regression with Regularization, Polynomial Regression, Random Forest Regression, and XGBoost Regression.

Linear regression we used as an interpretable baseline, polynomial regression we used to test for nonlinear relationships, random forest regression we used as it was robust to outliers and could handle nonlinear relationships, and XGBoost regression we used as it traditionally has the best predictive power for structured data.

Before we selected our final models, we standardized all our predictors to make the coefficients more interpretable and insensitive to feature scale. We also applied five-fold cross-validation to tune hyperparameters for each model without overfitting on the testing set. We used Root Mean Squared Error (RMSE) as the performance metric to evaluate performance in the validation set. RMSE is more sensitive to predictions that are far off and favors models that don't have predictions with large errors. Once we found the best hyperparameters, we randomly divided 80% of our data into a training set and 20% into a testing set, using a random state of 42 for reproducibility. In our testing set, we examined RMSE, Mean Absolute Error (MAE), and R-squared values.

Figures 7-9 show the performance of each hypertuned model on the testing set. We grouped the linear regression model that uses Singular Value Decomposition (SVD) and the regularized regression model that uses Stochastic Gradient Descent (SGD), as they had nearly identical performances. We also calculated a baseline model that predicts the mean AUDIT-C score of 3.08 for each patient, regardless of their predictors, for comparison.

Our XGBoost model performed the best in every performance metric, followed closely by linear regression and then random forest. The polynomial regression model did the worst by a fair margin, but all were significantly better than the baseline model. Our best model, the XGBoost, compared to our baseline had an RMSE of 2.043 vs 2.210, MAE of 1.554 vs 1.691, and an R-squared of 0.150 vs 0.00. This is about a 7.56% reduction in RMSE and 8.10% decrease in MAE.

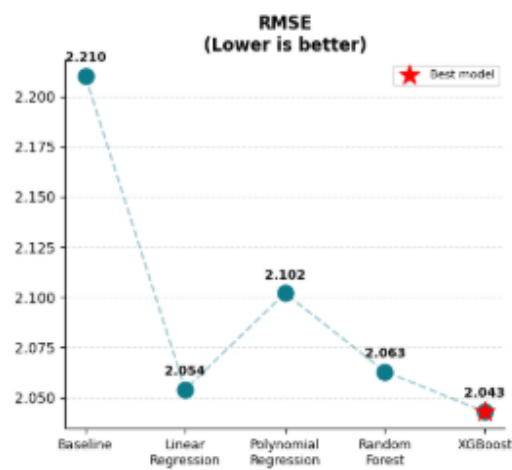


Figure 7: RMSE of each Model on the Testing Set

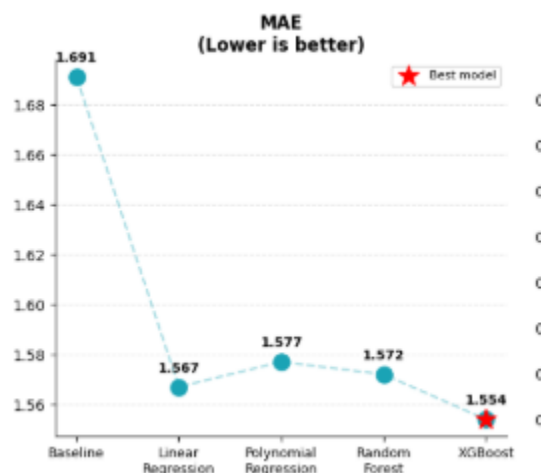


Figure 8: MAE of each Model on the Testing Set

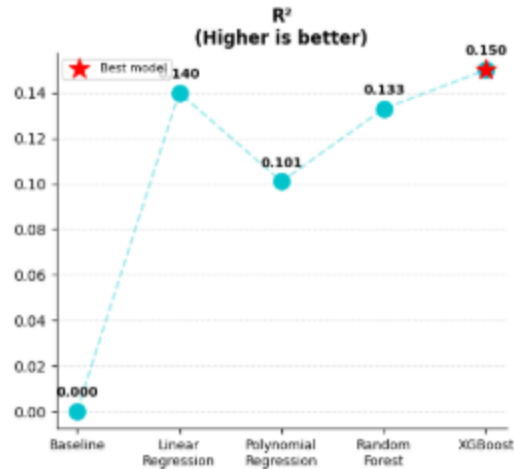


Figure 9: R-squared of each Model on the Testing Set

Our best model seems to have moderate to low predictive power, but it is significantly better than the naive baseline model. There are a few reasons why the predictive power may be low: we may have chosen bad predictors, patients may underreport their alcohol consumption and AUDIT-C score due to recall error, embarrassment, or social desirability bias, and it's difficult to predict a patient's exact score, so classification models that predict a patient being in a range of scores may perform better.

Feature importance was examined for the best-performing XGBoost model using both gain and SHAP values, as can be seen in Figures 10 and 11. Gain measures how much a feature improves tree splits when it is used, while SHAP values estimate the average contribution of each feature to individual predictions. SHAP shows how much the individual feature's value shifts the predicted AUDIT-C score compared to if the feature had the mean value.

In the gain-based ranking, the most important features were sex assigned at birth as male, current smoker status, reporting no recreational drug use, alcoholism medication use, household income less than \$10,000, being married, anxiety, some college education, AST, and liver disease. The gain values were highest for male sex at birth (1467.1), current smoker status (379.0), no recreational drug use (283.8), and alcoholism medication use (158.2), suggesting that these variables helped the model make its strongest splits. Features with the lowest gain included several smaller income and recreational drug-use categories, such as income between \$150,000 and \$200,000, income between \$50,000 and \$75,000, and less frequent substance-use categories. The model seems to be mainly splitting on the use of non-alcoholic substances and sex to reduce RMSE by the largest amount.

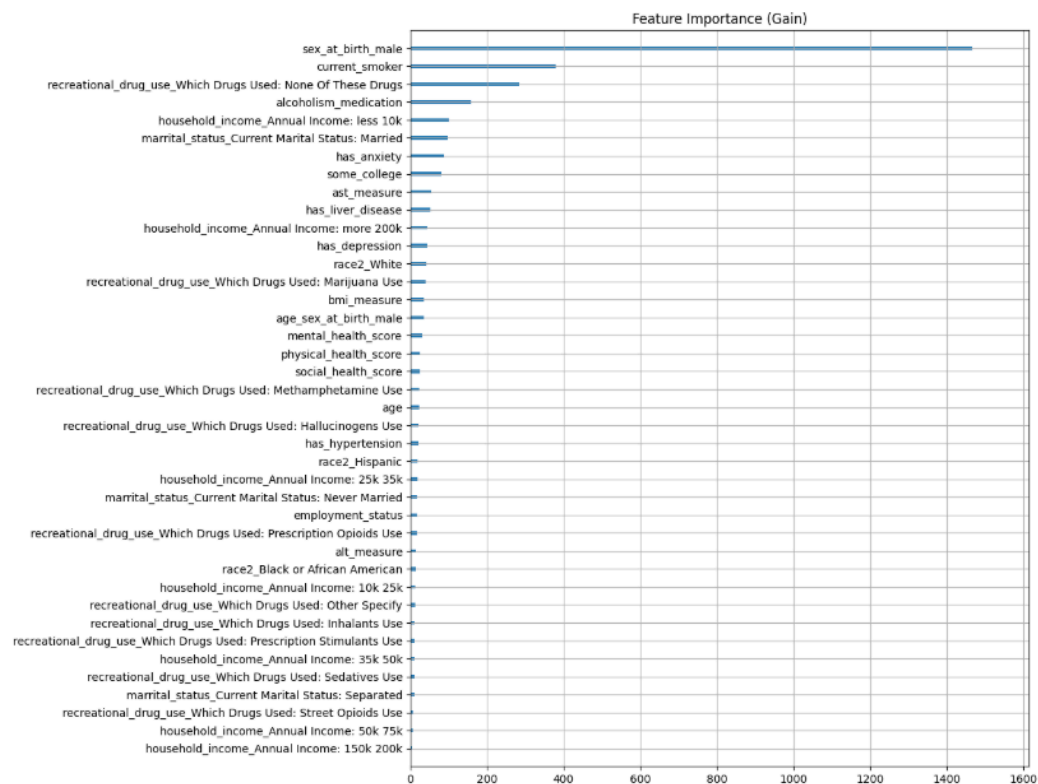


Figure 10: Feature Importance for XGBoost Model using Gain

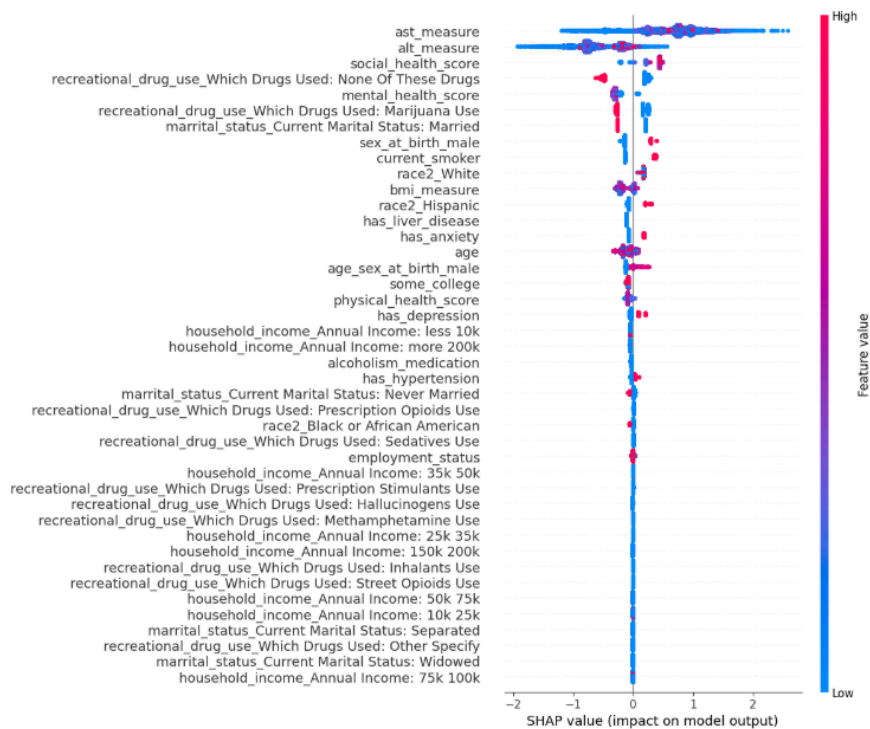


Figure 11: Feature Importance for XGBoost Model using SHAP Value Colored by Feature Value

The SHAP results gave a somewhat different interpretation of importance. AST and ALT had the largest average SHAP impact, followed by social health score, no recreational drug use, mental health score, marijuana use, marital status, sex assigned at birth, current smoker status, race, BMI, Hispanic ethnicity, liver disease, anxiety, age, and the age-by-sex interaction. This difference is useful because gain highlights features that create strong tree splits, while SHAP shows which features most consistently changed predicted AUDIT-C scores across patients. SHAP also shows directionality and the direction each feature is shifting AUDIT-C scores, as well as a quantitative measure of how much each feature value is shifting predictions. The SHAP plot has AST and ALT measures as the most impactful, as well as substance use, marital status, and mental health. Sex at birth, while important, is not by far and away the most important feature, as it was in the gain plot.

Overall, the feature-importance results suggest that alcohol consumption intensity was associated with a mix of demographic factors, smoking and substance-use behavior, liver-related lab measures, socioeconomic status, and mental/social health. However, the AST and ALT findings should be interpreted cautiously because these lab values were strongly correlated with each other in the correlation heatmap, so their individual importance may partly overlap.

Discussion:

This study found that machine learning models could predict AUDIT-C alcohol consumption intensity better than a naive mean-only baseline, but the overall predictive power remained modest. XGBoost performed best across the test-set metrics, with an RMSE of 2.043, MAE of 1.554, and R-squared of 0.150. This means the model captured some meaningful patterns in the data, but it did not explain most of the variation in AUDIT-C scores. In a real-world clinical or public health setting, this type of model may be more useful as a screening or risk-stratification tool than as a precise score prediction tool.

The most consistent predictors were not from one single domain. Demographic variables such as sex assigned at birth, behavioral variables such as current smoking and recreational drug-use responses, socioeconomic variables such as income and marital status, and clinical variables such as AST, ALT, liver disease, anxiety, and alcoholism medication use all contributed to model predictions. This supports the idea that alcohol consumption risk is multidimensional and connected to behavioral, social, and health-related factors. The feature-importance findings also align with clinical expectations because liver enzymes, smoking, substance-use history, and mental health conditions are commonly related to alcohol-use risk.

A key strength of this project is that it used a large All of Us cohort and combined several data domains, including demographics, surveys, measurements, conditions, and medications. The two-year observation window helped ensure that predictors were temporally close to the AUDIT-C survey responses. The analysis also compared multiple model types, used cross-validation for model tuning, evaluated results on a held-out test set, and interpreted the final model using both gain and SHAP values instead of relying only on prediction metrics.

There are also important limitations. First, AUDIT-C is self-reported, so patients may underreport drinking because of recall error, embarrassment, or social desirability bias. Second, the final complete-case cohort decreased from 262,087 eligible patients to 31,174 patients after requiring all predictors, mainly because measurement data were sparse. This may introduce selection bias if patients with complete lab and survey data differ from those excluded. Third, patients who reported never drinking in the past year were not included because they did not answer all three AUDIT-C questions, so the final cohort does not include AUDIT-C scores of 0. Fourth, the All of Us cohort is a volunteer research cohort and may not fully represent the general U.S. population. The final cohort was also majority female and majority White, which may affect generalizability across demographic groups.

Several modeling limitations should also be considered. AUDIT-C is an ordinal, bounded score from 0 to 12, but we modeled it using regression. This allowed us to test continuous prediction, but classification into risk groups may perform better and may be easier to apply clinically. Some predictors were also correlated, especially AST and ALT, anxiety and depression, and the three self-reported health measures. These correlations can make feature importance harder to interpret because related variables may share predictive signal. Finally, this was an observational analysis, so the model identifies associations rather than causal relationships.

Future work should test classification models that predict clinically meaningful AUDIT-C risk groups, such as low-risk versus hazardous drinking. This is what has traditionally been done in literature to predict alcohol consumption levels, achieving higher levels of success than our regression models. It's easier to predict a bin of scores for a patient than an individual score. It would also be useful to compare complete-case modeling with imputation-based approaches, include additional longitudinal predictors, and evaluate fairness across sex, race, ethnicity, and age groups. External validation of similar results on another cohort would be needed before using the model in practice, to ensure generalizability.

Conclusion:

Using All of Us data, this project showed that AUDIT-C alcohol consumption intensity can be predicted better than a baseline model by using a combination of demographic, survey, laboratory measurements, conditions, and medication features. XGBoost was the best-performing model, but its modest R-squared shows that exact AUDIT-C score prediction remains challenging. Our models' most significant features for elevated AUDIT-C scores, specifically high AST, low ALT, recreational drug and tobacco use, and lower mental health scores, are critical indicators for clinical consideration. While these factors can help physicians identify patients at a higher risk for alcohol misuse, it is important to note that no single variable is diagnostic of alcoholism in isolation. The results suggest that future work should focus on clinically meaningful risk classification, improved handling of missing data, and validation across more representative populations.

Team Contribution Statement:

Nick: Performed data cleaning to get AUDIT-C scores, index date, and observation window for patients, and got all relevant predictors cleaned into one dataframe for machine learning. Also wrote the introduction and methods section on the progress report, presentation, and final report.

Prathibha: focused on preprocessing and feature engineering, including cleaning survey responses, handling missing data, and transforming categorical variables into model-ready features.

Sushil: conducted exploratory data analysis to understand key patterns in alcohol consumption and predictor variables, and created visualizations to support our findings.

Ramcharan: worked on building and evaluating machine learning models, comparing different algorithms and tuning them to improve prediction performance.

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